YASHRAJ RASTOGI

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PROFESSIONAL SUMMARY

B.Tech Computer Science undergraduate (2024–2028) specializing in full-stack web development, AI integration, and scalable system architecture. Top 2% National Finalist (40,000+ participants) at the India AI Impact Buildathon and Top 10 Finalist at CodeRush Hackathon. Proficient in Next.js, React, Firebase, and PyTorch, building real-time, AI-driven applications with Google Gemini. Led campus tech initiatives as a Google Student Ambassador (GID: 3263) and Technical Head.

TECHNICAL SKILLS

Languages:	Java, JavaScript, TypeScript, Python, C
Frontend:	React.js (v18/v19), Next.js, Tailwind CSS v4, HTML5, Progressive Web Apps (PWA)
Backend:	Node.js, Express.js, Django, FastAPI, Cloud Functions (Serverless), Twilio API
AI / ML:	Google Gemini AI (Function Calling), PyTorch, Hugging Face, Wav2Vec 2.0, Librosa, MediaPipe
DB & Cloud:	MongoDB, Firebase (Firestore, Auth, Rules), Vercel, Hugging Face Spaces
Tools:	Git, GitHub, Figma, Canva, Vite, Postman
Concepts:	Full-Stack Development, REST API Design, System Architecture, Explainable AI (XAI), Role-Based Access Control

PROJECTS

Crisis OS — Real-Time Emergency Management PWA *2026*

React 18 · TypeScript · Firebase · Google Gemini AI · PWA · Tailwind CSS | GDG Solution Challenge 2026 — Team Leader
Live Demo · GitHub

- Architected a real-time PWA supporting five user roles (Admin, Manager, Staff, Guest, Responder) with role-based access, built for intense concurrent operational load.
- Built a live dashboard using Firestore real-time listeners, letting managers track guest safety status (Safe / Need Help / Critical) by floor and zone without refreshing.
- Integrated Google Gemini AI into the reporting pipeline to automatically convert raw field reports into severity-ranked action checklists and targeted broadcast drafts.
- Designed a secure serverless backend with Node.js Cloud Functions for automated role assignments and strict Firestore Security Rules ensuring complete data isolation across properties.

FinBuddy — AI Self-Checkout & Merchant Copilot *2026*

React · Vite · Tailwind CSS · Firebase (Auth, Firestore, Cloud Functions) · Google Gemini AI · Twilio | Hack-e-Awadh — Paytm PS-02 (Merchant Growth AI) Live Demo · GitHub

- Built an AI self-checkout: customers scan products (barcode or voice) to get an itemized bill and an exact-amount payment QR/link, tested via a payment simulator.
- Developed a conversational merchant copilot using Google Gemini that turns checkout transaction data into plain-language insights and growth recommendations in Hindi, Hinglish and English.
- Architected the backend with the React client on Firestore/Auth and Cloud Functions handling payments, inventory and Gemini calls; integrated Twilio for WhatsApp receipts.

RevGuard — AI Recovery for Failed Recurring Payments *2026*

Google Gemini AI · Firebase · Cloud Functions · React | Razorpay AI Buildathon — AI Revenue Recovery Track (Solo) Live Demo · GitHub

- Built solo an AI-powered recovery system for failed recurring payments and UPI mandate auto-debits, covering one complete loop: Ingest, Diagnose, Decide, Act, Audit and Measure.
- Diagnosed the cause of each failed debit and decided the best recovery action per transaction, recording every automated action in an audit trail for full traceability.
- Added a measurement stage tracking recovery outcomes so the impact on recovered revenue can be evaluated end to end.

NyayaPath — Multilingual Notice-to-Action Platform *2026*

React · Tailwind CSS · Firebase Auth · Google Gemini AI · Web Speech API | AI for Civic & Legal Empowerment Hackathon (Problem Statement 3) Live Demo · GitHub

- Developed a platform that converts uploaded government notices into plain-language English/Hindi explanations, extracting authority, deadline and notice type and generating an action plan.
- Implemented a single Gemini call with response_schema validation and a needs_review flag that escalates urgent or low-confidence notices (e.g. deadlines within 72 hours) to legal-aid reviewers.
- Built a guided RTI generator with real filing details (PIO addressee, Section 6(1) declaration, fee instructions) and PDF export.

VoiceGuard — AI Voice Detection REST API *2026*

Python · FastAPI · PyTorch · Hugging Face · Wav2Vec 2.0 (XLSR-53) | Top 2% — India AI Impact Buildathon (40,000+ participants) Live Demo · GitHub

- Engineered a hybrid AI detection pipeline combining a fine-tuned Wav2Vec 2.0 (XLSR-53) transformer with audio forensic algorithms to identify synthetic voice anomalies and acoustic artifacts.
- Built a high-performance REST API using FastAPI and Uvicorn capable of real-time, low-latency audio analysis via Base64-encoded payload submissions.
- Implemented Explainable AI (XAI) features parsing diagnostic model metrics (e.g., artificial spectral smoothness) into human-readable summaries for non-technical reviewers.

AcadVault — Academic Management Portal *2025*

React.js · Tailwind CSS · Firebase | Smart India Hackathon (SIH 2025) — Internal Round Qualifier Live Demo · GitHub

- Built a comprehensive academic management portal for students, faculty, and administrators, qualifying through internal SIH collegiate rounds.
- Implemented role-based access control enabling students to manage profiles, faculty to verify achievements, and administrators to update academic records and generate portfolios.

LEADERSHIP & EXPERIENCE

Google Student Ambassador · Google India · GID: 3263 *May 2026 – Sep 2026*

- Organized monthly hands-on technical workshops campus-wide advocating for Google AI tools including Gemini and NotebookLM, managing end-to-end event logistics.
- Provided 1-on-1 technical consultations to student founders, identifying operational bottlenecks and integrating tools like NotebookLM and Gemini Canvas to accelerate project development.

Technical Head · Unstop Igniters Club SRMCEM *Apr 2026 – Present*

- Lead technical teams in website development and digital infrastructure initiatives supporting the club's events and online presence.

Co-Lead & Technical Lead · AlgoZenith Chapter SRMCEM *Oct 2025 – Present*

- Organized the 'Vibe Coding Contest' hackathon and 'Algo-Arena' DSA competition, managing logistics and facilitating competitive programming for students.
- Coordinated technical workshops led by industry professionals to enhance practical skills within the collegiate community.

EDUCATION

B.Tech in Computer Science and Engineering *2024 – 2028*
Shri Ramswaroop Memorial College of Engineering & Management, Lucknow · *CGPA: 7.5 / 10*

Intermediate Science (12th Grade) · City Convent School · 90.25% (9.02 CGPA) *2022 – 2023*

High School (10th Grade) · City Convent School · 84% (8.4 CGPA) *2020 – 2021*

CERTIFICATIONS & AWARDS

- Top 2% National Finalist — India AI Impact Buildathon, HCL GUVI (40,000+ participants) [Feb 2026] View Credential
- Top 10 Finalist — CodeRush Hackathon [2026]
- Top 10 — Hack Nerds Hackathon (offline, PW IOI)
- Team Leader — GDG Solution Challenge 2026 (Crisis OS) [2026]
- Team Leader / Participant — GDG Solution Challenge 2025 [2025] View Credential
- HeLa Crossroads Certification — HeLa Labs [2024] View Credential
- Web Development Certification — Sololearn [2024] View Credential
- Java Certification — Sololearn [2024] View Credential